Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech

Semester: V

Name of the Course: Computer Networks-I

Course Code: TCS 514

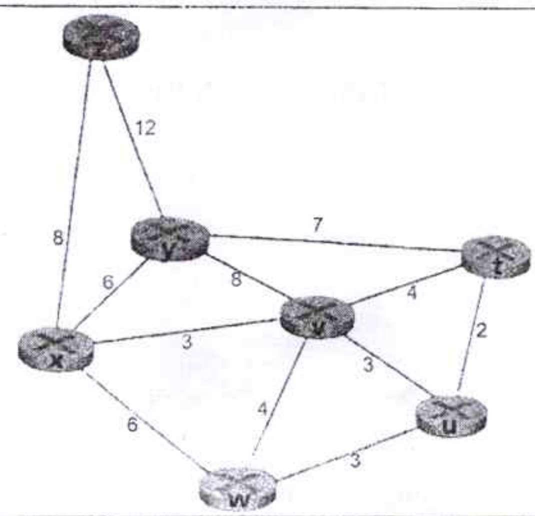
Time: 3 Hours

Maximum Marks: 100

Note:

- i. **All questions are compulsory.**
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What are the benefits of using a layered architecture in networking? How does the TCP/IP protocol stack differ from the OSI model?	CO1
(b)	Do signals actually travel faster in fibre optic cables? State the working principle of fibre optics and mention its key applications in the business world.	CO1
(c)	Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps. (i) Assuming no other traffic in the network, what is the throughput for the file transfer? (ii) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B?	CO1
Q2		
(a)	Differentiate between persistent and non-persistent HTTP connections.	CO2
(b)	Consider the 5-bit generator, G = 10011, and suppose that D has the value 1010101010. What is the value of R?	CO2
(c)	What is the principal difference between virtual circuits and datagram networks? Give an example.	CO2
Q3		
(a)	What is a logically centralized control plane? In such architectures, are the control plane and data plane located on the same device or on separate devices? Explain briefly.	CO4
(b)	Describe the Domain Name System (DNS) and its key components.	CO3
(c)	Explain the structure and elements of a TCP header in detail.	CO3
Q4		
(a)	Consider the below network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from x to all network nodes.	CO5 CO5

		CO4
(b)	Explain how email is sent and retrieved using standard Internet protocols. Include a discussion on the role of SMTP in email transmission, the structure of a typical email message, and a comparison between POP3 and IMAP as email access protocols.	
(c)	Draw and explain the FSM of the rdt2.1 sender and the rdt2.1 receiver. How does it differ from the rdt2.2 sender and the rdt2.2 receiver?	
Q5		
(a)	Explain why mapping between IP address and MAC address is required in a network. Why can't the MAC address itself serve the purpose of an IP address? Also, state whether ARP can be used in networks other than Ethernet.	CO5
(b)	Explain how CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) helps in reducing data collisions in wireless networks. Describe the steps involved in the process.	CO6
(c)	In a CSMA/CD network operating at 10 Mbps, a node experiences its **fifth collision**. (i) What is the probability that the node selects $\mathbf{K = 4}$ during exponential backoff? (ii) What is the corresponding **delay in seconds** for this K value?	CO6